Honors Mathematics II
Functions of a Single Variable

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Welcome to MATH186 !

- Please read the Course Description, which has been uploaded to the Files section on the Canvas course site.
- This course will be offered online.
- I am always available for a personal conversation; online office hours will be announced on Canvas. You can also make an appointment by email or write to me with any questions. My email is horst@sjtu.edu.cn
- The Teaching Assistants for this course will provide recitation classes, office hours, and help with grading.

Office Hours / Piazza

In addition to being available in office hours, the TAs and I will be answering course-related questions on Piazza. Please also create an account such that your name in pinyin is visible.

It is possible to send private messages on Piazza, but most messages should be public so that everyone can see them and the responses or respond themselves. Feel free to answer other students' questions!

Please do not post anonymously unless you have a good reason. Don't be shy!

Please post messages in English only.

Here is the sign-up link:

https://piazza.com/sjtu.org/fall2025/math1860j

Coursework

- There will be weekly coursework (assignments) throughout the term.
- You will be randomly assigned into ***assignment groups*** of three students; you are expected to collaborate within each group and hand in a single, common solution paper to each coursework.
- Each group must achieve 60% of the total coursework points by the end of the term in order to obtain a passing grade for the course. However, the assignment points have no effect on the course grade.
- Please hand in your coursework on time, by the date given on each set of course work. Late work will not be accepted unless you come to me personally and I find your explanation for the lateness acceptable.
- You can be deducted up to 10% of the awarded marks for an assignment if you fail to write neatly and legibly.
- Further details can be found in the course description.

Use of AI and Other Sources; Honor Code Policy

When faced with a particularly difficult problem, you may want to refer to external sources: textbooks, websites such as Wikipedia or various AI chat bots. Here are a few guidelines:

- External sources may treat a similar sounding subject matter at a much more advanced or a much simpler level than this course. This means that explanations you find are much more complicated or far too simple to help you. For example, when looking up the "induction axiom" you may find many high-school level explanations that are not sufficient for our problems; on the other hand, Wikipedia contains a lot of information relating to formal logic that is far beyond what we are discussing here.
- If you do use any outside sources to help you solve a homework problem, you are not allowed to just copy the solution; this is considered a violation of the Honor Code.

Use of AI and Other Sources; Honor Code Policy

- The correct way of using outside sources is to understand the contents of your source and then to write in your own words and without referring back to the source the solution of the problem. Your solution should differ in style significantly from the published solution. If you are not sure whether you are incorporating too much material from your source in your solutions, then you must cite the source that you used.
- You may and are required to collaborate freely with other students in your assignment group. However, you may not communicate at all about concrete coursework with students from other groups. However, discussing general questions regarding the lecture contents with any other student is of course fine and encouraged.

  Do not show or explain your solutions to any student outside your assignment group.

Use of AI and Other Sources; Honor Code Policy

In this course, the following actions are examples of violations of the Honor Code ("another student" means a student outside your assignment group):

- Showing another student your written solution to a problem.
- Sending a screenshot of your solution via QQ, email or other means to another student.
- Showing another student the written solution of a third student; distributing some student's solution to other students.
- Viewing another student's written solution.
- Copying your solution in electronic form (LaTeX source, PDF, JPG image etc.) to the computer hardware (flash drive, hard disk etc.) of another student. Having another student's solution in electronic form on your computer hardware.

If you have any questions regarding the application of the Honor Code, please contact me or any of the TAs.

Grading Policy

- The grade will be composed of the course work and the exams as follows:
  - Participation: 5 points
  - Course Outcome Quizzes: 5 points
  - Midterm exam: 45 points
  - Final exam: 45 points
- The actual grading scale will usually be based on the top approximately 6-12% of students receiving a grade of A+, with the following grades determined by (mostly) fixed point increments.
- Apart from this normalization, the grade distribution is up to you! If (for example) all students obtain many points in the exams, I am happy to see everyone receive a grade of A. Students are primarily evaluated with respect to a fixed point scale, not with respect to each other.

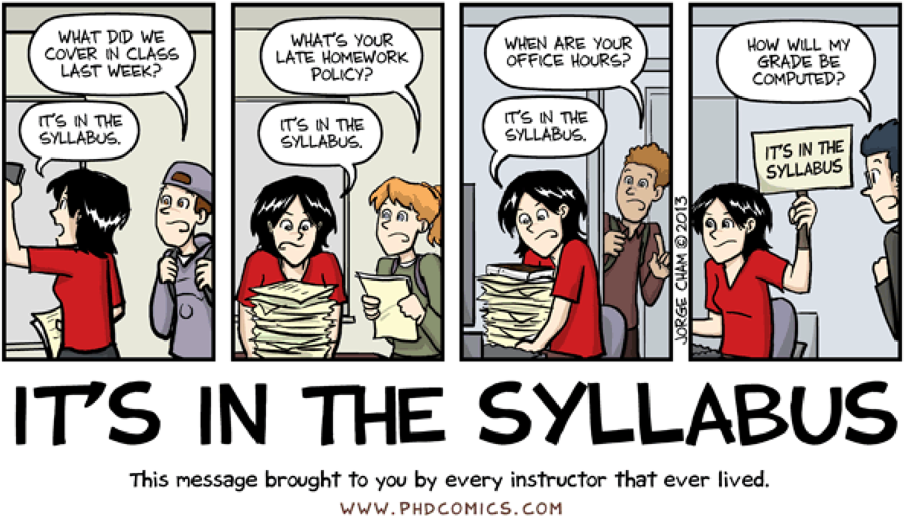
Course Grade Example – MATH1860J in Fall 2023

Points	Grade	No. of students	% of students	Percentile
0	F	1	0.7%	0.0%
40	D	4	2.7%	0.7%
45	C-	4	2.7%	3.4%
50	C	10	6.7%	6.0%
55	C+	10	6.7%	12.8%
60	B-	12	8.1%	19.5%
65	B	14	9.4%	27.5%
70	B+	23	15.4%	36.9%
75	A-	35	23.5%	52.3%
80	A	21	14.1%	75.8%
85	A+	15	10.1%	89.9%

Course Grade Example – MATH1860J in Fall 2024

Points	Grade	No. of students	% of students	Percentile
0	F	0	0.0%	0.0%
35	D	0	0.0%	0.0%
40	C-	0	0.0%	0.0%
45	C	0	0.0%	0.0%
50	C+	7	4.5%	0.0%
55	B-	8	5.1%	4.5%
60	B	25	16.0%	9.6%
65	B+	36	23.1%	25.6%
70	A-	44	28.2%	48.7%
75	A	23	14.7%	76.9%
80	A+	13	8.3%	91.7%

More Info: Syllabus (a.k.a. Course Description)



What is Mathematics?

Mathematics is about certainty.

Mathematics is about modeling.

Mathematics is about thinking.

Course Focus

> *Please forget everything you have learned in school; you haven't learned it.*
>
> *Please keep in mind everywhere the corresponding portions of your school work; you haven't actually forgotten them.*

Edmund Landau, Foundations of Analysis, 1929

The course will focus on functions of one variable, in particular

- Foundations: Set Theory, Logic, Numbers, Algebraic Structures.
- Functions, Limits and Continuity,
- Differentiation
- Integration

Course Topics: Foundations

1. Basic Concepts in Logic
2. Basic Concepts in Set Theory
3. The Natural Numbers and Mathematical Induction
4. The Rational Numbers
5. Roots, Bounds & Real Numbers
6. Complex Numbers

Course Topics: Functions, Convergence and Continuity

7. Functions and Maps

8. Sequences

9. Real Functions

10. Limits of Functions

11. Continuous Functions

Course Topics: Derivatives, Vector Spaces and Series

12. Differentiation of Real Functions
13. Properties of Differentiable Real Functions
14. Vector Spaces
15. Sequences of Real Functions
16. Series
17. Real and Complex Power Series
18. The Exponential Function
19. The Trigonometric Functions

Course Topics: Integrals and Applications

20. Notions of Integration

21. Practical Integration

22. Applications of Integration

Suggested Reading I

[1] K. Jänich, ***Linear Algebra*** (Undergraduate Texts in Mathematics). Springer-Verlag New York, 1994.
DOI: 10.1007/978-1-4612-4298-7.

[2] H. H. Sohrab, ***Basic Real Analysis***. Birkhäuser Basel, 2003.
DOI: 10.1007/978-0-8176-8232-3.

[3] L. H. Loomis and S. Sternberg, ***Advanced Calculus, Revised Edition***. Jones and Bartlett, London, 1990.
URL: https://people.math.harvard.edu/~shlomo/.

[4] M. A. Pons, ***Real Analysis for the Undergraduate*** (Undergraduate Texts in Mathematics). Springer, New York, 2014.
DOI: 10.1007/978-1-4614-9638-0.

[5] O. Hijab, ***Introduction to Calculus and Classical Analysis*** (Undergraduate Texts in Mathematics), 4th ed. Springer, Cham, 2016.
DOI: 10.1007/978-3-319-28400-2.

Suggested Reading II

[6] A. Browder, ***Mathematical Analysis*** (Undergraduate Texts in Mathematics). Springer, New York, NY, 1996.
DOI: 10.1007/978-1-4612-0715-3.

[7] J. M. Howie, ***Real Analysis*** (Springer Undergraduate Mathematics Series). Springer, London, 2001.
DOI: 10.1007/978-1-4471-0341-7.

[8] M. Spivak, ***Calculus***, 4th ed. Publish or Perish Press, Houston, TX, 2008.
URL:
https://archive.org/details/spivak-m.-calculus-2008/.

Notation

The notation used in these notes should be fairly well-known from school; some symbols will be defined explicitly to make sure that they are understood unambiguously (for example, the summation symbol $\sum$).

A particular convention needs some explanation: when an equality is a definition (and not a statement that can be proven), the equal symbol will sometimes have a colon added on the side of the defined quantity. For example,

$$A \cup B := \{x \colon x \in A \text{ or } x \in B\}$$

indicates that that the union $A \cup B$ is being defined through this equality. Another example might involve

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2} =: a_n,$$

indicating that the number a_n is defined to be $n(n+1)/2$.

Notation

On the other hand, it would be wrong to write

$$1 + 2 + \cdots + n := \frac{n(n+1)}{2}$$

because the objects on the left and the right are already defined and the equality is a statement that can be proven; it does not define anything.

Part I

Foundations

Propositional Logic

Reference [3, Sections 0.1-0.4]

Propositional logic is the field of manipulatiing statements. A ***statement*** (also called a ***proposition***) is anything that is labeled unambigiuously as being either ***true*** or ***false***.

We do not define what the words “statement”, “true” or “false” mean. Instead we deal with objects that every reader will agree are either a “true statement” or a “false statement.”

We will generally not use examples from the “real world” as statements. The reason is that objects in the real world are much to loosely defined for the application of strict logic to make any sense. For example, the statement “It is raining.” may be considered true by some people (“Yes, raindrops are falling out of the sky.”) while at the same time false by others (“No, it is merely drizzling.”) Furthermore, important information is missing (Where is it raining? When is it raining?).

Statements

1.1.1. Examples.

- "$3 > 2$" is a ***true statement***.
- "$x^3 > 0$" is not a statement, because we can not decide whether it is true or not (what is "x," actually?).
- "the cube of any integer is strictly positive" is a ***false statement***.

The last example can be written using a ***statement variable*** x:

- "For any number x, $x^3 > 0$"

The first part of the statement is a ***quantifier*** ("for any number x"), while the second part is called a ***statement frame*** or ***predicate*** ("$x^3 > 0$").

A statement frame becomes a statement (which can then be either true or false) when the variable takes on a specific value; for example, $1^3 > 0$ is a true statement and $0^3 > 0$ is a false statement.

We will treat quantifiers and predicates in more detail later.

Working with Statements

We will denote statements by capital letters such as $A, B, C, \ldots$ and statement frames by symbols such as $A(x)$ or $B(x, y, z)$ etc.

1.1.2. Examples.

- A: 4 is an even number.
- B : $2 > 3$.
- $A(n)$: $1 + 2 + 3 + \ldots + n = n(n + 1)/2$.

We will now introduce ***logical operations*** on statements. One or more logical operations acting on one or more statements will yield a ***compound statement***.

Negation and Truth Tables

The simplest possible type of operation is a ***unary operation***, i.e., it takes a statement A and returns a statement B.

1.1.3. Definition. Let A be a statement. Then the ***negation*** of A, written as $\neg A$ and pronounced "not A" is the statement that is true if A is false and false if A is true.

1.1.4. Example. If A is the statement $A\colon 2 > 3$, then the negation of A is $\neg A\colon 2 \not> 3$.

We can describe the action of the unary operation $\neg$ through the following table:

A	$\neg A$
T	F
F	T

If A is true (T), then $\neg A$ is false (F) and vice versa. Such a table is called a ***truth table***.

Conjunction

The next simplest type of operations on statements are ***binary operations***. The have two statements as arguments and return a single statement, called a ***compound statement***, whose truth or falsehood depends on the truth or falsehood of the original two statements.

1.1.5. Definition. Let A and B be two statements. Then we define the ***conjunction*** of A and B, written $A \wedge B$, by the following truth table:

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

The conjunction $A \wedge B$ is spoken "A and B." It is true only if both A and B are true, false otherwise.

Disjunction

1.1.6. Definition. Let A and B be two statements. Then we define the ***disjunction*** of A and B, written $A \vee B$, by the following truth table:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

The disjunction $A \vee B$ is spoken "A or B." It is true only if either A or B is true, false otherwise.

1.1.7. Example.

- Let A: $2 > 0$ and $B: 1 + 1 = 1$. Then $A \wedge B$ is false and $A \vee B$ is true.
- Let A be a statement. Then the compound statement "$A \vee (\neg A)$" is always true, and "$A \wedge (\neg A)$" is always false.

Tautology and Proofs using Truth Tables

How do we prove that "$A \vee (\neg A)$" is an always true statement? We are claiming that $A \vee (\neg A)$ will be a true statement, regardless of whether the statement A is true or not. To prove this, we go through all possibilities using a truth table:

A	$\neg A$	$A \vee (\neg A)$
T	F	T
F	T	T

Since the column for $A \vee (\neg A)$ only lists T for "true," we see that $A \vee (\neg A)$ is always true. A logical operator that always yields true (compound) statements is called a ***tautology***.

Contradiction

Correspondingly, the truth table for $A \wedge (\neg A)$ is:

A	$\neg A$	$A \wedge (\neg A)$
T	F	F
F	T	F

A logical operator that always yields false compound statements is called a ***contradiction***.

Implication

1.1.8. Definition. Let A and B be two statements. Then we define the ***implication*** of B and A, written $A \Rightarrow B$, by the following truth table:

A	B	$A \Rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

We read "$A \Rightarrow B$" as "A implies B," "if A, then B" or "A only if B".

The fact that F $\Rightarrow$ T is a true statement may seem a little strange. Before we go into the reason for this, note that Definition 1.1.8 is a ***definition***! That means, we don't have to justify it in any way, there is nothing to prove, it is just a notation for a certain logical operator. Hence, we could leave it at that and give no further explanation. However, it would be useful and appropriate to ***motivate*** why we are defining the implication in this way.

Implication

To illustrate why the implication is defined as it is, it is useful to look at a specific example:

$$\text{For all integers } x, \qquad x > 0 \Rightarrow x^3 \geq 0 \tag{1.1.1}$$

It is hoped that the reader will agree that this should be a true statement. In order to verify this, we would need to verify the truth of the statement obtained from the predicate "$x > 0 \Rightarrow x^3 \geq 0$" for all integers x.

We shall see later that an integer can be either strictly positive, strictly negative or zero (cf. the trichotomy law on Slide 78).

- If x is a strictly positive number, both statements are true, so we have $\mathrm{T} \Rightarrow \mathrm{T}$.
- If x is a strictly negative number, both statements are false, so we have $\mathrm{F} \Rightarrow \mathrm{F}$.
- If x is zero, the first statement is false but the second statement is true, so we have $\mathrm{F} \Rightarrow \mathrm{T}$.

Equivalence (Logical Operator)

Hence, if we want (1.1.1) to be a true statement, then we need to define $A \Rightarrow B$ as being true in all of the three cases above. We next introduce yet another, important, logical operator.

1.1.9. Definition. Let A and B be two statements. Then we define the ***equivalence*** of A and B, written $A \Leftrightarrow B$, by the following truth table:

A	B	$A \Leftrightarrow B$
T	T	T
T	F	F
F	T	F
F	F	T

We read "$A \Leftrightarrow B$" as "A is equivalent to B" or "A if and only if B". Some textbooks abbreviate "if and only if" by "iff."

If A and B are both true or both false, then they are equivalent. Otherwise, they are not equivalent. In propositional logic, "equivalence" is the closest thing to the "equality" of arithmetic.

Logical Equivalence of Compound Statements

On the one hand, logical equivalence is strange; two statements A and B do not need to have anything to do with each other to be equivalent. For example, the statements "$2 > 0$" and "$100 - 1 = 99$" are both true, so they are equivalent.

1.1.10. Definition. Two compound statements A and B are said to be ***logically equivalent*** if their equivalence is a tautology, i.e., $A \Leftrightarrow B$ is always true. We then write $A \equiv B$.

1.1.11. Do It Yourself. The two ***de Morgan rules*** are the tautologies

$$\neg(A \vee B) \Leftrightarrow (\neg A) \wedge (\neg B), \qquad \neg(A \wedge B) \Leftrightarrow (\neg A) \vee (\neg B).$$

To indicate that they are tautologies, we write

$$\neg(A \vee B) \equiv (\neg A) \wedge (\neg B), \qquad \neg(A \wedge B) \equiv (\neg A) \vee (\neg B).$$

Contraposition

An important logical equivalence is the ***contrapositive***,

$$(A \Rightarrow B) \equiv (\neg B \Rightarrow \neg A).$$

For example, for any number x, the statement "$x > 0 \Rightarrow x^3 > 0$" is equivalent to "$x^3 \not> 0 \Rightarrow x \not> 0$." This principle is used in proofs by contradiction.

We prove the contrapositive using a truth table:

A	B	$\neg A$	$\neg B$	$\neg B \Rightarrow \neg A$	$A \Rightarrow B$	$(A \Rightarrow B) \Leftrightarrow (\neg B \Rightarrow \neg A)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Rye Whiskey

The following song is an old Western-style song, called “Rye Whiskey” and performed by Tex Ritter in the 1930’s and 1940’s.

If the ocean was whiskey and I was a duck,
I’d swim to the bottom and never come up.

But the ocean ain’t whiskey, and I ain’t no duck,
So I’ll play jack-of-diamonds and trust to my luck.

For it’s whiskey, rye whiskey, rye whiskey I cry.
If I don’t get rye whiskey I surely will die.

The lyrics make sense (at least as much as song lyrics generally do).

Rye Whiskey

One can use de Morgan's rules and the contrapositive to re-write the song lyrics as follows

If I never reach bottom or sometimes come up,
Then the ocean's not whiskey, or I'm not a duck.

But my luck can't be trusted, or the cards I'll not buck,
So the ocean is whiskey or I am a duck.

For it's whiskey, rye whiskey, rye whiskey I cry.
If my death is uncertain, then I get whiskey (rye).

These lyrics seem to be logically equivalent to the original song, but are just humorous nonsense. This again illustrates clearly why it is futile to apply mathematical logic to everyday language.

This example is due to (clickable link) W. P. Cooke, The American Mathematical Monthly, Vol. 76, No. 9 (Nov., 1969), p. 1051.

Logical Quantifiers

In the previous examples we have used predicates $A(x)$ with the words "for all x." This is an instance of a ***logical quantifier*** that indicates for which x a predicate $A(x)$ is to be evaluated to a statement.

In order to use quantifiers properly, we clearly need a universe of objects x which we can insert into $A(x)$ (a ***domain*** for $A(x)$). This leads us immediately to the definition of a ***set***. We will discuss set theory in detail later. For the moment it is sufficient for us to view a set as a "collection of objects" and assume that the following sets are known:

- the set of natural numbers $\mathbb{N}$ (which includes the number 0),
- the set of integers $\mathbb{Z}$,
- the set of real numbers $\mathbb{R}$,
- the empty set $\emptyset$ (also written $\varnothing$ or $\{\}$) that does not contain any objects.

If M is a set containing x, we write $x \in M$ and call x an ***element*** of M.

Logical Quantifiers

There are two types of quantifiers:

- the ***universal quantifier***, denoted by the symbol $\forall$, read as "for all" and
- the ***existential quantifier***, denoted by $\exists$, read as "there exists."

1.1.12. Definition. Let M be a set and $A(x)$ be a predicate. Then we define the quantifier $\forall$ by

$$\mathop{\forall}_{x\in M} A(x) \quad \Leftrightarrow \quad A(x) \text{ is true for all } x \in M$$

We define the quantifier $\exists$ by

$$\mathop{\exists}_{x\in M} A(x) \quad \Leftrightarrow \quad A(x) \text{ is true for at least one } x \in M$$

We may also write $\forall x \in M\colon A(x)$ instead of $\mathop{\forall}_{x\in M} A(x)$ and similarly for $\exists$.

Logical Quantifiers

We may also state the domain before making the statements, as in the following example.

1.1.13. Examples. Let x be a real number. Then

- $\forall x\colon x > 0 \Rightarrow x^3 > 0$ is a true statement;
- $\forall x\colon x > 0 \Leftrightarrow x^2 > 0$ is a false statement;
- $\exists x\colon x > 0 \Leftrightarrow x^2 > 0$ is a true statement.

Sometimes mathematicians put a quantifier at the end of a statement frame; this is known as a ***hanging quantifier***. Such a hanging quantifier will be interpreted as being located just before the statement frame:

$$\exists y\colon y + x^2 > 0 \qquad\qquad \forall x$$

is equivalent to $\exists y \forall x\colon y + x^2 > 0$.

Contraposition and Negation of Quantifiers

We do not actually need the quantifier $\exists$ since

$$\begin{aligned}\underset{x\in M}{\exists} A(x) \quad &\Leftrightarrow \quad A(x) \text{ is true for at least one } x \in M \\ &\Leftrightarrow \quad A(x) \text{ is not false for all } x \in M \\ &\Leftrightarrow \quad \neg \underset{x\in M}{\forall} (\neg A(x)) \qquad (1.1.2)\end{aligned}$$

The equivalence (1.1.2) is called ***contraposition of quanifiers***. It implies that the negation of $\exists x \in M\colon A(x)$ is equivalent to $\forall x \in M\colon \neg A(x)$. For example,

$$\neg \big(\exists x \in \mathbb{R}\colon x^2 < 0\big) \qquad \Leftrightarrow \qquad \forall x \in \mathbb{R}\colon x^2 \not< 0.$$

Conversely,

$$\neg (\forall x \in M\colon A(x)) \qquad \Leftrightarrow \qquad \exists x \in M\colon \neg A(x).$$

Vacuous Truth

If the domain of the universal quantifier $\forall$ is the empty set $M = \varnothing$, then the statement $\forall x \in M \colon A(x)$ is defined to be true regardless of the predicate $A(x)$. It is then said that $A(x)$ is ***vacuously true***.

1.1.14. Example. Let M be the set of real numbers x such that $x = x + 1$. Then the statement

$$\forall_{x \in M} \; x > x$$

is true.

This convention reflects the philosophy that a universal statement is true unless there is a counterexample to prove it false. While this may seem a strange point of view, it proves useful in practice.

This is similar to saying that "All pink elephants can fly." is a true statement, because it is impossible to find a pink elephant that can't fly.

Nesting Quantifiers

We can also treat predicates with more than one variable as shown in the following example.

1.1.15. Examples. In the following examples, x, y are taken from the real numbers.

- $\forall x \forall y\colon x^2 + y^2 - 2xy \geq 0$ is equivalent to $\forall y \forall x\colon x^2 + y^2 - 2xy \geq 0$. Therefore, one often writes $\forall x, y\colon x^2 + y^2 - 2xy \geq 0$.
- $\exists x \exists y\colon x + y > 0$ is equivalent to $\exists y \exists x\colon x + y > 0$, often abbreviated to $\exists x, y\colon x + y > 0$.
- $\forall x \exists y\colon x + y > 0$ is a true statement.
- $\exists x \forall y\colon x + y > 0$ is a false statement.

As is clear from these examples, the order of the quantifiers is important if they are different.

Naive Set Theory: Sets via Predicates

We want to be able to talk about "collections of objects"; however, we will be unable to strictly define what an "object" or a "collection" is (except that we also want any collection to qualify as an "object"). The problem with "naive" set theory is that any attempt to make a formal definition will lead to a contradiction - we will see an example of this later. However, for our practical purposes we can live with this, as we won't generally encounter these contradictions.

We indicate that an object (called an ***element***) x is part of a collection (called a ***set***) X by writing $x \in X$. We characterize the elements of a set X by some predicate P:

$$x \in X \qquad \Leftrightarrow \qquad P(x). \tag{1.2.1}$$

We write such a set X in the form $X = \{x : P(x)\}$.

Notation for Sets

We define the empty set $\varnothing := \{x : x \neq x\}$. The empty set has no elements, because the predicate $x \neq x$ is never true.

We may also use the notation $X = \{x_1, x_2, \dots, x_n\}$ to denote a set. In this case, X is understood to be the set

$$X = \{x : (x = x_1) \vee (x = x_2) \vee \cdots \vee (x = x_n)\}.$$

We will frequently use the convention

$$\{x \in A : P(x)\} = \{x : x \in A \wedge P(x)\}$$

1.2.1. Example. The set of even positive integers is

$$\{n \in \mathbb{N} : \underset{k \in \mathbb{Z}}{\exists}\, n = 2k\}$$

Subsets and Equality of Sets

If every object $x \in X$ is also an element of a set Y, we say that X is a ***subset*** of Y, writing $X \subset Y$; in other words,

$$X \subset Y \qquad \Leftrightarrow \qquad \forall x \in X : x \in Y.$$

We say that $X = Y$ if and only if $X \subset Y$ and $Y \subset X$.

We say that X is a ***proper subset*** of Y if $X \subset Y$ but $X \neq Y$. In that case we write $X \subsetneq Y$.

Some authors write $\subseteq$ for $\subset$ and $\subset$ for $\subsetneq$. Pay attention to the convention used when referring to literature.

Examples of Sets and Subsets

1.2.2. Examples.

1. For any set X, $\varnothing \subset X$. Since $\varnothing$ does not contain any elements, the domain of the statement $\forall y \in \varnothing \colon y \in X$ is empty. Therefore, it is vacuously true and hence $\varnothing \subset X$.
2. Consider the set $A = \{a, b, c\}$ where a, b, c are arbitrary objects, for example, numbers. The set

$$B = \{a, b, a, b, c, c\}$$

is equal to A, because it satisfies $A \subset B$ and $B \subset A$ as follows:

$$x \in A \quad \Leftrightarrow \quad (x = a) \vee (x = b) \vee (x = c) \quad \Leftrightarrow \quad x \in B.$$

Therefore, neither order nor repetition of the elements affects the contents of a set.

If $C = \{a, b\}$, then $C \subset A$ and in fact $C \subsetneqq A$. Setting $D = \{b, c\}$ we have $D \subsetneqq A$ but $C \not\subset D$ and $D \not\subset C$.

Power Set and Cardinality

If a set X has a finite number of elements (a formal statement of what "finite" means here will be given later, in Definition 2.1.9), we define the ***cardinality*** of X to be this number, denoted by $\#X$, $|X|$ or $\operatorname{card} X$.

We define the ***power set***

$$\mathscr{P}(M) := \{A \colon A \subset M\}.$$

Here the elements of the set $\mathscr{P}(M)$ are themselves sets; $\mathscr{P}(M)$ is the "set of all subsets of M." Therefore, the statements

$$A \subset M \qquad \text{and} \qquad A \in \mathscr{P}(M)$$

are equivalent.

1.2.3. Example. The power set of $\{a, b, c\}$ is

$$\mathscr{P}(\{a, b, c\}) = \big\{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\big\}.$$

The cardinality of $\{a, b, c\}$ is 3, the cardinality of the power set is $|\mathscr{P}(\{a, b, c\})| = 8$.

Operations on Sets

If $A = \{x\colon P_1(x)\}$, $B = \{x\colon P_2(x)\}$ we define the ***union***, ***intersection*** and ***difference*** of A and B by

$$A \cup B := \{x\colon P_1(x) \lor P_2(x)\}, \qquad A \cap B := \{x\colon P_1(x) \land P_2(x)\},$$
$$A \setminus B := \{x\colon P_1(x) \land (\neg P_2(x))\}.$$

Let $A \subset M$. We then define the ***complement*** of A by

$$A^{\mathrm{c}} := M \setminus A.$$

If $A \cap B = \emptyset$, we say that the sets A and B are ***disjoint***.

Occasionally, the notation $A - B$ is used for $A \setminus B$ and A^{c} is sometimes denoted by $\overline{A}$.

1.2.4. Example. Let $A = \{a, b, c\}$ and $B = \{c, d\}$. Then

$$A \cup B = \{a, b, c, d\}, \qquad A \cap B = \{c\}, \qquad A \setminus B = \{a, b\}.$$

Operations on Sets

The laws for logical equivalencies immediately lead to several rules for set operations. For example, de Morgan's laws (see Example 1.1.11) for $\wedge$ and $\vee$ imply

- $(A \cup B)^c = A^c \cap B^c$,
- $(A \cap B)^c = A^c \cup B^c$,

Other such rules are, for example,

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$
- $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$
- $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Some of these will be proved in the recitation class and the assignments.

Operations on Sets

Occasionally we will need the following notation for the union and intersection of a finite number $n \in \mathbb{N}$ of sets:

$$\bigcup_{k=0}^{n} A_k := A_0 \cup A_1 \cup A_2 \cup \cdots \cup A_n,$$
$$\bigcap_{k=0}^{n} A_k := A_0 \cap A_1 \cap A_2 \cap \cdots \cap A_n.$$

This notation even extends to $n = \infty$, but needs to be properly defined:

$$x \in \bigcup_{k=0}^{\infty} A_k \quad :\Leftrightarrow \quad \underset{k \in \mathbb{N}}{\exists} x \in A_k,$$
$$x \in \bigcap_{k=0}^{\infty} A_k \quad :\Leftrightarrow \quad \underset{k \in \mathbb{N}}{\forall} x \in A_k.$$

Operations on Sets

In particular,

$$\bigcap_{k=0}^{\infty} A_k \subset \bigcup_{k=0}^{\infty} A_k.$$

1.2.5. Example. Let $A_k = \{0, 1, 2, \dots, k\}$ for $k \in \mathbb{N}$. Then

$$\bigcup_{k=0}^{\infty} A_k = \mathbb{N}, \qquad \bigcap_{k=0}^{\infty} A_k = \{0\}.$$

To see the first statement, note that $\mathbb{N} \subset \bigcup_{k=0}^{\infty} A_k$ since $x \in \mathbb{N}$ implies $x \in A_x$ implies $x \in \bigcup_{k=0}^{\infty} A_k$. Furthermore, $\bigcup_{k=0}^{\infty} A_k \subset \mathbb{N}$ since $x \in \bigcup_{k=0}^{\infty} A_k$ implies $x \in A_k$ for some $k \in \mathbb{N}$ implies $x \in \mathbb{N}$.

For the second statement, note that $\bigcap_{k=0}^{\infty} A_k \subset \mathbb{N}$. Now $0 \in A_k$ for all $k \in \mathbb{N}$. Thus $\{0\} \subset \bigcap_{k=0}^{\infty} A_k$. On the other hand, for any $x \in \mathbb{N} \setminus \{0\}$ we have $x \notin A_{x-1}$ whence $x \notin \bigcap_{k=0}^{\infty} A_k$.

Ordered Pairs

References [3, Section 0.6], see also [8, Appendix to Chapter 3]

A set does not contain any information about the order of its elements, e.g.,

$$\{a, b\} = \{b, a\}.$$

Thus, there is no such a thing as the “first element of a set”. However, sometimes it is convenient or necessary to have such an ordering. This is achieved by defining an ***ordered pair***, denoted by

$$(a, b)$$

and having the property that

$$(a, b) = (c, d) \qquad \Leftrightarrow \qquad (a = c) \wedge (b = d). \tag{1.2.2}$$

Cartesian Product of Sets

One way to implement (1.2.2) is to define

$$(a, b) := \{\{a\}, \{a, b\}\} \tag{1.2.3}$$

1.2.6. Do It Yourself. Check that if (a, b) and (c, d) (*mutatis mutandis*) are defined by (1.2.3), then property (1.2.2) holds.

If A, B are sets and $a \in A$, $b \in B$, then we denote the set of all ordered pairs by

$$A \times B := \{(a, b) \colon a \in A,\ b \in B\}. \tag{1.2.4}$$

$A \times B$ is called the ***cartesian product*** of A and B.

If we take the cartesian product of a set with itself, we may abbreviate it using exponents, e.g.,

$$\mathbb{N}^2 := \mathbb{N} \times \mathbb{N}.$$

Cartesian Product of Sets

1.2.7. Do It Yourself. Generalize (1.2.3) to define an ***ordered triple*** (a, b, c) or, more generally, an ordered ***n-tuple*** $(a_1, \dots, a_n)$ for $n \in \mathbb{N} \setminus \{0\}$.

The n-fold cartesian product $A_1 \times \cdots \times A_n$ of sets A_k, $k = 1, \dots, n$, is then defined analogously to (1.2.4).

Problems in Naive Set Theory

If one simply views sets as arbitrary "collections" and allows a set to contain arbitrary objects, including other sets, then fundamental problems arise. We first illustrate this by an analogy:

Suppose a library contains not only books but also catalogs of books, i.e., books listing other books. For example, there might be a catalog listing all mathematics books in the library, a catalog listing all history books, etc. Suppose that there are so many catalogs, that you are asked to create catalogs of catalogs, i.e., catalogs listing other catalogs. In particular, you are asked to create the following:

(i) A catalog of all catalogs in the library. This catalog lists all catalogs contained in the library, so it must of course also list itself.

(ii) A catalog of all catalogs that list themselves. Does this catalog also list itself?

(iii) A catalog of all catalogs that do not list themselves. Does this catalog also list itself?

The Russel Antinomy

In the previous analogy, we can view “catalogs” as “sets” and being “listed in a catalog” as “being an element of a set”. Then we have

(i) The set of all sets must have itself as an element.

(ii) The set of all sets that have themselves as elements may or may not contain itself. (This may be decidable by adding some rule to set theory.)

(iii) It is not decidable whether the “set of all sets that do not have themselves as elements” has itself as an element.

The Russel Antinomy

Formally, this paradox is known as the ***Russel antinomy:***

1.2.8. Russel Antinomy. The predicate $P(x)\colon x \notin x$ does not define a set $A = \{x \colon P(x)\}$.

Proof.
If $A = \{x \colon x \notin x\}$ were a set, then we should be able to decide for any set y whether $y \in A$ or $y \notin A$. We show that for $y = A$ this is not possible because either assumption leads to a contradiction:

(i) Assume $A \in A$. Then $P(A)$ by (1.2.1), i.e., $A \notin A$. $\lightning$

(ii) Assume $A \notin A$. Then $\neg P(A)$ by (1.2.1), therefore $A \in A$. $\lightning$

Since we cannot decide whether $A \in A$ or $A \notin A$, A can not be a set. $\square$

The Russel Antinomy

There are several examples in classical literature and philosophy of the Russel antimony and other"paradoxes":

1. A person says: "This sentence is a lie." Is he lying or telling the truth? (An example of a sentence which is not a logical statement, since a truth value can not be assigned to it.)
2. In a mountain village, there is a barber. Some villagers shave themselves (always) while the others never shave themselves. The barber shaves those and only those villagers that never shave themselves. Who shaves the barber?

   (A pseudo-illustration of the Russel antimony. For details,see J. Raclavsky, ***The Barber Paradox: On its Paradoxicality and its Relationship to Russell's Paradox***, Prolegomena **13**:2, 2014, Pages 269–278). Available here: https://philpapers.org/rec/RACTBP.)

Russel Antinomy

We will simply ignore the existence of such contradictions and build on naive set theory. There are further paradoxes (antinomies) in naive set theory, such as ***Cantor's paradox*** and the ***Burali-Forti paradox***. All of these are resolved if naive set theory is replaced by a ***modern axiomatic set theory*** such as ***Zermelo-Fraenkel set theory.***

Further Information:

- ***Set Theory***, Stanford Encyclopedia of Philosophy, http://plato.stanford.edu/entries/set-theory/
- P.R. Halmos, ***Naive Set Theory***, Available here: http://link.springer.com/book/10.1007/978-1-4757-1645-0
- T. Jech, ***Set Theory: The Third Millennium Edition, Revised and Expanded***, Available here: http://link.springer.com/book/10.1007/3-540-44761-X

The Natural Numbers

We denote the set of natural numbers by $\mathbb{N}$, i.e.,

$$\mathbb{N} = \{0, 1, 2, 3, 4, \ldots\}.$$

The natural numbers can be ***constructed*** rigorously from set theory, but we will not do so here.

We assume that the familiar operations of addition and multiplication are known for the natural numbers, i.e., for any two natural numbers $a, b \in \mathbb{N}$ we can define the natural number $c = a + b \in \mathbb{N}$ called the ***sum*** of a and b. The addition has the following properties (here $a, b, c \in \mathbb{N}$):

1. $a + (b + c) = (a + b) + c$ (***Associativity***)
2. $a + 0 = 0 + a = a$ (***Existence of a neutral element***)
3. $a + b = b + a$ (***Commutativity***)

Multiplication

Similarly, we can define ***multiplication***, where $a \cdot b \in \mathbb{N}$ is called the ***product*** of a and b. We have the following properties (here $a, b, c \in \mathbb{N}$):

1. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ (***Associativity***)
2. $a \cdot 1 = 1 \cdot a = a$ (***Existence of a neutral element***)
3. $a \cdot b = b \cdot a$ (***Commutativity***)

We also have a property that essentially states that addition and multiplication are ***consistent***,

$$a \cdot (b + c) = a \cdot b + a \cdot c. \qquad \text{(\textbf{\textit{Distributivity}})}$$

Note that we are not able to define subtraction or division for all natural numbers.

Notation for Addition and Multiplication

For any numbers $a_1, a_2, \dots, a_n$ we define the notation

$$a_1 + a_2 + \cdots + a_n =: \sum_{j=1}^{n} a_j =: \sum_{1 \le j \le n} a_j$$

and

$$a_1 \cdot a_2 \cdots a_n =: \prod_{j=1}^{n} a_j =: \prod_{1 \le j \le n} a_j.$$

(We will use this notation for rational and real numbers a_k; see the following sections.)

For $n \in \mathbb{N}$ we define

$$0! := 1 \qquad \text{and} \qquad n! := n \cdot (n-1)! \quad \text{for } n > 0.$$

This is an example of a ***recursive definition***.

Powers and Ordering

We are also able to define the exponential, “intuitively” written as

$$a^n = \underbrace{a \cdot a \cdot \ldots \cdot a}_{n \text{ times}} \qquad\qquad \text{for } a, n \in \mathbb{N}$$

by setting $a^0 := 1$ and $a^n := a \cdot a^{n-1}$ (another recursive definition). We note that

$$a^{m+n} = a^m \cdot a^n \qquad \text{and} \qquad (a^m)^n = a^{m \cdot n} \qquad \text{for } a, m, n \in \mathbb{N}.$$

We can imbue the natural numbers with an ***ordering***, denoted by $<$. We say that $a < b$ if there exists some $c \in \mathbb{N}^* := \mathbb{N} \setminus \{0\}$ such that $b = a + c$. We write $a > b$ if $b < a$.

We define the symbol $\le$ to mean “$<$ or $=$”, so

$$a \le b \qquad\qquad \Leftrightarrow \qquad\qquad \big(a < b \vee a = b\big).$$

We write $a \ge b$ if $b \le a$.

Mathematical Induction

Often one wants to show that some statement frame $A(n)$ is true for all $n \in \mathbb{N}$ with $n \geq n_0$ for some $n_0 \in \mathbb{N}$. Mathematical induction works by establishing two statements:

(I) $A(n_0)$ is true.

(II) $A(n+1)$ is true whenever $A(n)$ is true for $n \geq n_0$, i.e.,

$$\underset{\substack{n \in \mathbb{N} \\ n \geq n_0}}{\forall} \big(A(n) \Rightarrow A(n+1)\big)$$

Note that (II) does not make a statement on the situation when $A(n)$ is false; it is permitted for $A(n+1)$ to be true even if $A(n)$ is false.

The ***principle of mathematical induction*** now claims that $A(n)$ is true for all $n \geq n_0$ if (I) and (II) are true. Essentially, this principle is part of the definition of $\mathbb{N}$.

Binomial Formula

In practice, one first establishes $A(n_0)$, then shows that $A(n+1)$ is true, assuming that $A(n)$ holds.

As an example, we will prove the binomial formula,

$$A(n): \qquad \underset{a,b\in\mathbb{R}}{\forall} (a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k},$$

for all $n \in \mathbb{N}$. Here

$$\binom{n}{k} := \frac{n!}{k!(n-k)!} \qquad\qquad \text{for } n, k \in \mathbb{N} \text{ and } k \le n.$$

1.3.1. Do It Yourself. Show that

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k} \qquad \text{for } n, k \in \mathbb{N} \text{ and } 1 \le k \le n. \qquad (1.3.1)$$

Binomial Formula

We start by showing that $A(0)$ is true. We see that

$$(a+b)^0 = 1, \qquad \text{and} \qquad \sum_{k=0}^{0} \binom{0}{k} a^k b^{0-k} = \binom{0}{0} a^0 b^{0-0} = 1.$$

Hence $A(0)$ is true, since both sides of the equation are equal to unity.

We now consider $A(n+1)$:

$$(a+b)^{n+1} = (a+b)(a+b)^n = (a+b)\sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k} \tag{1.3.2}$$

We have inserted the statement $A(n)$ in (1.3.2), since we want to show that $A(n+1)$ is true ***under the condition that $A(n)$ is true***.

Binomial Formula

Applying some algebraic manipulations, we obtain

$$\begin{aligned}(a+b)^{n+1} &= (a+b)\sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k}\\ &= \sum_{k=0}^{n}\binom{n}{k}a^{k+1}b^{n-k} + \sum_{k=0}^{n}\binom{n}{k}a^k b^{n+1-k}\\ &= \sum_{j=1}^{n+1}\binom{n}{j-1}a^j b^{n+1-j} + \sum_{k=0}^{n}\binom{n}{k}a^k b^{n+1-k}\end{aligned}$$

where we have set $j = k + 1$.

Binomial Formula

We can now rename the index back to k and split off some terms of the sums:

$$\begin{aligned}(a+b)^{n+1} &= a^{n+1} + \sum_{k=1}^{n} \binom{n}{k-1} a^k b^{n+1-k} + b^{n+1} + \sum_{k=1}^{n} \binom{n}{k} a^k b^{n+1-k} \\ &= a^{n+1} + b^{n+1} + \sum_{k=1}^{n} \left(\binom{n}{k-1} + \binom{n}{k} \right) a^k b^{n+1-k}\end{aligned}$$

Now we apply (1.3.1):

$$\begin{aligned}(a+b)^{n+1} &= a^{n+1} + b^{n+1} + \sum_{k=1}^{n} \binom{n+1}{k} a^k b^{n+1-k} \\ &= \sum_{k=0}^{n+1} \binom{n+1}{k} a^k b^{n+1-k},\end{aligned}$$

which is just $A(n+1)$. Hence the binomial formula holds for all $n \in \mathbb{N}$.

Basic Properties of Numbers - Addition

Reference [8, Chapter 1]

We assume that the set of rational numbers (fractions)

$$\mathbb{Q} = \left\{ \frac{p}{q} \colon p, q \in \mathbb{Z} \ \wedge \ q \neq 0 \right\}$$

is known and has been introduced from set theory. We recapitulate the basic structural properties of the rational numbers. For addition we have

associativity $$\forall_{a,b,c \in \mathbb{Q}} \quad a + (b + c) = (a + b) + c \qquad \textbf{(P1)}$$

neutral element $$\exists_{0 \in \mathbb{Q}} \ \forall_{a \in \mathbb{Q}} \quad a + 0 = 0 + a = a \qquad \textbf{(P2)}$$

inverse element $$\forall_{a \in \mathbb{Q}} \ \exists_{-a \in \mathbb{Q}} \quad (-a) + a = a + (-a) = 0 \qquad \textbf{(P3)}$$

commutativity $$\forall_{a,b \in \mathbb{Q}} \quad a + b = b + a. \qquad \textbf{(P4)}$$

Multiplication

For multiplication, we have a similar set of properties

associativity	$\underset{a,b,c\in\mathbb{Q}}{\forall} \quad a\cdot(b\cdot c)=(a\cdot b)\cdot c,$	**(P5)**
neutral element	$\underset{\substack{1\in\mathbb{Q}\\1\neq 0}}{\exists}\ \underset{a\in\mathbb{Q}}{\forall} \quad a\cdot 1=1\cdot a=a,$	**(P6)**
inverse element	$\underset{\substack{a\in\mathbb{Q}\\a\neq 0}}{\forall}\ \underset{a^{-1}\in\mathbb{Q}}{\exists} \quad a\cdot a^{-1}=a^{-1}\cdot a=1,$	**(P7)**
commutativity	$\underset{a,b\in\mathbb{Q}}{\forall} \quad a\cdot b=b\cdot a.$	**(P8)**

With these properties we can prove that if $a\neq 0$ and $a\cdot b=a\cdot c$, then $b=c$.

Combining Addition and Multiplication

These twice four properties for addition and multiplication are basically independent of each other (the only connection was that $0 \neq 1$).

The following property ensures that the two operations "work well together",

$$\text{distributivity} \qquad \forall_{a,b,c \in \mathbb{Q}} \quad a \cdot (b + c) = a \cdot b + a \cdot c. \qquad \textbf{(P9)}$$

It is this relation that allows us to prove that $a - b = b - a$ only if $a = b$, and it is also the law of distributivity that we employ to do elementary multiplications. Without (P9) we cannot prove that if $a \cdot b = 0$, then either $a = 0$ or $b = 0$, for example.

The Positive Rational Numbers

We assume that we know what a strictly positive rational number is, i.e., that there exists a set P with the property that for every number a one and only one of the following holds:

(i) $a = 0$,

(ii) $a \in P$,

(iii) $-a \in P$.

This is known as the ***trichotomy law*** **(P10)** for the rational numbers. We also assume that the set of positive numbers is closed under addition and multiplication:

$$\text{If } a \text{ and } b \text{ are in } P, \text{ then } a + b \text{ is in } P, \qquad \textbf{(P11)}$$

$$\text{If } a \text{ and } b \text{ are in } P, \text{ then } a \cdot b \text{ is in } P. \qquad \textbf{(P12)}$$

The Ordering Relation

We then write

$$P = \{p \in \mathbb{Q} \colon p > 0\}.$$

so that $p > 0$ is defined by $p \in P$. Furthermore, for two rational numbers we can then define

$$a > b \quad :\Leftrightarrow \quad a - b > 0 \quad \Leftrightarrow \quad a - b \in P.$$

We say that we have an ***ordering relation*** on the set of rational numbers. We also write

$$a \geq b \quad :\Leftrightarrow \quad a > b \ \vee \ a = b.$$

The existence of the set P hence allows us to define what $>$ means and the properties (P11) and (P12) ensure that the ordering relation is compatible with multiplication and addition.

The Absolute Value

We define the ***absolute value*** (sometimes called the ***modulus***) $|a|$ of a rational number a by

$$|a| = \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

An important basic result is the ***triangle equality***:

1.4.1. Triangle Inequality. For all rational numbers $a, b \in \mathbb{Q}$, we have

$$|a + b| \leq |a| + |b|. \tag{1.4.1}$$

Proof.

We prove the triangle inequality by considering all possible cases for a and b according to the trichotomy law (P10):

- $a = 0,\ b \in \mathbb{Q}$. If $a = 0$ we have $a + b = b$ and $|a| = a = 0$, so $|a| + |b| = |b|$. Both sides of (1.4.1) are then equal to $|b|$, so the inequality holds.

The Triangle Inequality

Proof.

We prove the triangle inequality by considering all possible cases for a and b according to the trichotomy law:

- $a > 0,\ b > 0$. In this case, $a + b > 0$ by (P11) so $|a + b| = a + b$. Furthermore, $|a| = a$ and $|b| = b$, so both sides of (1.4.1) are equal to $a + b$.
- $a < 0,\ b < 0$. In this case, $a + b < 0$ by (P11) (why?) so $|a + b| = -a - b$. Furthermore, $|a| = -a$ and $|b| = -b$, so both sides of (1.4.1) are equal to $-a - b$.
- $a < 0,\ b > 0,\ a + b \geq 0$. In this case, $|a| + |b| = b - a$ and $|a + b| = b + a$. Since $a < 0$ and $-a > 0$ we have

  $$|a + b| = b + a < b + 0 = b < b + (-a) = b - a = |a| + |b|$$

  and the inequality holds.

The case $a < 0,\ b > 0,\ a + b < 0$ is left to the reader! $\square$

The Triangle Inequality

As a corollary of the triangle inequality we obtain the ***reverse triangle inequality***:

$$|a+b| \geq \big||a| - |b|\big| \qquad \text{for all } a, b \in \mathbb{Q}. \tag{1.4.2}$$

For the proof, it is sufficient to apply the triangle inequality as follows:

$$|b| = |a+b-a| \leq |a+b| + |a|.$$

Rearranging gives

$$|a+b| \geq |b| - |a|.$$

Interchanging a and b then yields (1.4.2).

The Rational Numbers are Insufficient

While the rational numbers are closed with respect to addition and multiplication, and furthermore are large enough to include “inverse elements” $-a$ and a^{-1} for these operations, they are still “incomplete” in more than one sense:

1. Algebraically: there is no inverse element for operation of squaring. Given $b \in \mathbb{Q}$, it may not be possible to find $a \in \mathbb{Q}$ such that $a^2 = b$.
2. Analytically: Given a sequence of rational numbers that come closer and closer to each other, there may not be a “limiting value” in $\mathbb{Q}$. (We will discuss this later.)
3. Set theoretically: Given a bounded subset of $\mathbb{Q}$, there may not be a least upper or greatest lower bound for this set.

We will discuss i) and iii) in more detail now, and come back to ii) later, when we construct a suitable expansion of the rational numbers.

The Square Root Problem

Reference Spivak, end of Chapter 2.

1.5.1. Theorem. There exists no number $a \in \mathbb{Q}$ such that $a^2 = 2$.

Proof.
Assume that $a = p/q \in \mathbb{Q}$ has the property that $(p/q)^2 = 2$, where $p, q \in \mathbb{N}$. We further assume that p and q have no common divisor greater than 1. Then

$$p^2 = 2q^2,$$

so p^2 is even. This implies that p is even, so $p = 2k$ for some $k \in \mathbb{N}$. But then

$$2q^2 = 4k^2 \qquad \Rightarrow \qquad q^2 = 2k^2 \text{ is even} \qquad \Rightarrow \qquad q \text{ is even.}$$

Thus p and q are both divisible by 2. $\lightning$ $\square$

Bounded Subsets of $\mathbb{Q}$

Reference [8, Beginning of Chapter 8]

1.5.2. Definition. We say that a set $U \subset \mathbb{Q}$ is ***bounded*** if there exists a constant $c \in \mathbb{Q}$ such that

$$|x| \leq c \qquad\qquad \text{for all } x \in U.$$

If this is not the case, we say that U is ***unbounded***.

Numbers c_1 and c_2 such that

$$c_1 \leq x \leq c_2 \qquad\qquad \text{for all } x \in U$$

are called ***lower*** and ***upper bounds*** for U, respectively.

We say that a set $U \in \mathbb{Q}$ is ***bounded above*** if there exists an upper bound for U, and ***bounded below*** if there exists a lower bound for U.

Bounded Subsets of $\mathbb{Q}$

1.5.3. Examples.

- The set $U = \{1/n \in \mathbb{Q} : n \in \mathbb{N}^*\}$ is bounded since $|x| \le 1$ for all $x \in U$. Furthermore, the number -1 is a lower bound and 1 is an upper bound. Of course, there are many more upper and lower bounds.
- The set $\mathbb{N} \subset \mathbb{Q}$ is bounded below (lower bound: 0), but not bounded above. It is unbounded.

Maxima and Minima of Sets

1.5.4. Definition. Let $U \subset \mathbb{Q}$ be a subset of the rational numbers. We say that a number $x_1 \in U$ is the ***minimum*** of U if

$$x_1 \le x, \qquad \text{for all } x \in U$$

and we write $x_1 =: \min U$.

Similarly, we say that $x_2 \in U$ is the ***maximum*** of U if

$$x_2 \ge x, \qquad \text{for all } x \in U$$

and write $x_2 =: \max U$.

A set that is not bounded below has no minimum, and a set that is not bounded above has no maximum. However, even a bounded set does not need to have a maximum or a minimum.

Greatest Lower and Least Upper Bounds of Sets

1.5.5. Definition. We say that an upper bound $c_2 \in \mathbb{Q}$ of a set $U \subset \mathbb{Q}$ is the ***least upper bound*** or ***supremum*** of U if no $c \in \mathbb{Q}$ with $c < c_2$ is an upper bound. We then write $c_2 =: \sup U$.

We say that a lower bound $c_1 \in \mathbb{Q}$ is the ***greatest lower bound*** or ***infimum*** of U if no $c \in \mathbb{Q}$ with $c > c_1$ is a lower bound. We then write $c_1 =: \inf U$.

1.5.6. Example. The set $U = \{1/n \in \mathbb{Q} \colon n \in \mathbb{N}^*\}$ has

$$\begin{aligned} &\inf U = 0, && \sup U = 1, \\ &\min U \text{ does not exist}, && \max U = 1. \end{aligned}$$

In general, if $\max U$ exists, then $\sup U = \max U$, and if $\min U$ exists, then $\inf U = \min U$.

Existence of Suprema and Infima

Not every bounded set in $\mathbb{Q}$ has an infimum or supremum: consider

$$U = \{x \in \mathbb{Q} \colon 2 < x^2 \leq 4 \wedge x > 0\}$$

It is clear that $\sup U = \max U = 2$, but the minimum and even the infimum do not exist (you will prove this in the assignments).

On the other hand, if we add a postulate to our previous axioms, we can widen the rational numbers to include solutions of $x^2 = 2$.

The Real Numbers

We define the set of real numbers $\mathbb{R}$ as the smallest extension of the rational numbers $\mathbb{Q}$ such that the following property holds:

> **(P13)** *If $A \subset \mathbb{R}$, $A \neq \emptyset$ is bounded above, then there exists a least upper bound for A in $\mathbb{R}$.*

We call all real numbers that are not rational ***irrational numbers***.

We will not immediately construct such a set of objects, but do so a little later in this lecture. For now, we stipulate that such a set exists, and we call it the set of ***real numbers*** $\mathbb{R}$. Furthermore, we can add and multiply elements of $\mathbb{R}$ and in general all properties (P1)-(P12) remain valid for elements of $\mathbb{R}$.

Note that (P13) is sufficient to guarantee the existence of greatest lower bounds; we need but consider $-A = \{x \in \mathbb{R} : -x \in A\}$ instead of A. The least upper bound of $-A$ will yield the negative of the greatest lower bound of A.

The Square Root Problem

We can now solve the square root problem:

1.5.7. Theorem. For every real number $x > 0$, there exists a unique positive real number y solving the equation $y^2 = x$.

Proof.
The proof is in two parts: existence and uniqueness, i.e., we show separately that i) there exists a solution and ii) the solution is unique. Often, it is easier to show uniqueness than existence. Also, the uniqueness proof may yield helpful results that can be used for the (generally more difficult) existence proof. Therefore, one often starts with proving the uniqueness.

Assume that there are two numbers $y_1 > y_2 > 0$ such that $y_1^2 = y_2^2 = x$. Then

$$0 = y_1^2 - y_2^2 = \underbrace{(y_1 - y_2)}_{>0}\underbrace{(y_1 + y_2)}_{>0} > 0 \lightning$$

The Square Root Problem

Proof (continued).
We now show the existence of a solution to the equation $y^2 = x$. Let $M := \{t \in \mathbb{R} : t > 0 \wedge t^2 > x\}$. Then M is non-empty and bounded from below, so by (P13) it has a greatest lower bound $y = \inf M$. We will establish that $y^2 = x$ by showing that $y^2 > x$ and $y^2 < x$ lead to contradictions. This strategy essentially uses the trichotomy law (P10).

Assume that $y^2 > x$, so $y > \frac{x}{y}$. Then

$$0 < \left(y - \frac{x}{y}\right)^2 = y^2 - 2y\frac{x}{y} + \left(\frac{x}{y}\right)^2,$$

so that

$$y^2 + \left(\frac{x}{y}\right)^2 > 2y\frac{x}{y} = 2x.$$

The Square Root Problem

Proof (continued).
Set $s := \frac{1}{2}(y + \frac{x}{y})$. Then

$$s^2 = \frac{1}{4}\left(y^2 + 2\frac{x}{y}y + \left(\frac{x}{y}\right)^2\right) > \frac{1}{4}(2x + 2x) = x$$

so $s \in M$. However,

$$s = \frac{1}{2}(y + \frac{x}{y}) < \frac{1}{2}(y + y) = y = \inf M \lightning$$

The proof that $y^2 < x$ leads to a contradiction is simpler and will be discussed in the assignments. $\square$

Roots and Exponents

With more work, it is possible to prove that the set of real numbers includes all positive solutions to $y^n = x$. Given $x \geq 0$, we hence define

$$y = \sqrt[n]{x}, \qquad\qquad n \in \mathbb{N} \setminus \{0\},$$

to be the unique number $y \geq 0$ such that $y^n = x$. We define the odd root of a negative number by setting

$$y := -\sqrt[2n+1]{-x}, \qquad\qquad \text{for } x < 0 \text{ and } n \in \mathbb{N}.$$

We recall that x^n for $n \in \mathbb{N}$ has been defined on Slide 68 (with $a \in \mathbb{N}$ replaced by $x \in \mathbb{R}$). For any $x \in \mathbb{R}$ for which the following makes sense, we define

$$x^{m/n} := \sqrt[n]{x^m}, \qquad\qquad m \in \mathbb{N},\ n \in \mathbb{N} \setminus \{0\},$$

and

$$x^{-q} := \frac{1}{x^q}, \qquad\qquad q \in \mathbb{Q}.$$

Subsets of the Real Numbers

We introduce some notation we will have occasion to use often later.

1.5.8. Definition. Let $a, b \in \mathbb{R}$ with $a \le b$. Then we define the following special subsets of $\mathbb{R}$, which we call ***intervals***:

$$\begin{aligned}
[a, b] &:= \{x \in \mathbb{R} \colon (a \le x) \wedge (x \le b)\} = \{x \in \mathbb{R} \colon a \le x \le b\}, \\
[a, b) &:= \{x \in \mathbb{R} \colon (a \le x) \wedge (x < b)\} = \{x \in \mathbb{R} \colon a \le x < b\}, \\
(a, b] &:= \{x \in \mathbb{R} \colon (a < x) \wedge (x \le b)\} = \{x \in \mathbb{R} \colon a < x \le b\}, \\
(a, b) &:= \{x \in \mathbb{R} \colon (a < x) \wedge (x < b)\} = \{x \in \mathbb{R} \colon a < x < b\}.
\end{aligned}$$

Furthermore, for any $a \in \mathbb{R}$ we set

$$\begin{aligned}
[a, \infty) &:= \{x \in \mathbb{R} \colon x \ge a\}, & (a, \infty) &:= \{x \in \mathbb{R} \colon x > a\}, \\
(-\infty, a] &:= \{x \in \mathbb{R} \colon x \le a)\}, & (-\infty, a) &:= \{x \in \mathbb{R} \colon x < a\}
\end{aligned}$$

Finally, we set $(-\infty, \infty) := \mathbb{R}$.

Subsets of the Real Numbers

1.5.9. Definition. We call an interval I

- ***open*** if $I = (a, b)$, $a \in \mathbb{R}$ or $a = -\infty$, $b \in \mathbb{R}$ or $b = \infty$;
- ***closed*** if $I = [a, b]$, $a, b \in \mathbb{R}$;
- ***half-open*** if is neither open or closed.

We often denote intervals by capital letters I or J. If we say "Let I be an open interval..." we mean "Let $I \subset \mathbb{R}$ be a set such that there exist numbers $a \in \mathbb{R}$ or $a = -\infty$ and $b \in \mathbb{R}$ or $b = \infty$ such that $I = (a, b)$..."

Subsets of the Real Numbers

More generally, we have the following classification of points with respect to a set:

1.5.10. Definition. Let $A \subset \mathbb{R}$ be any set.

(i) We call $x \in \mathbb{R}$ an ***interior point of A*** if there exists some $\varepsilon > 0$ such that the interval $(x - \varepsilon, x + \varepsilon) \subset A$. The set of interior points of A is denoted by $\operatorname{int} A$.

(ii) We call $x \in \mathbb{R}$ an ***exterior point of A*** if there exists some $\varepsilon > 0$ such that the interval $(x - \varepsilon, x + \varepsilon) \cap A = \emptyset$.

(iii) We call $x \in \mathbb{R}$ a ***boundary point of A*** if for every $\varepsilon > 0$ $(x - \varepsilon, x + \varepsilon) \cap A \neq \emptyset$ and $(x - \varepsilon, x + \varepsilon) \cap A^{\mathrm{c}} \neq \emptyset$. The set of boundary points of A is denoted by ∂A.

(iv) We call $x \in \mathbb{R}$ an ***accumulation point of A*** if for every $\varepsilon > 0$ the interval $(x - \varepsilon, x + \varepsilon) \cap A \setminus \{x\} \neq \emptyset$.

Subsets of the Real Numbers

1.5.11. Example. For the interval $A = [0, 1)$,

- $\operatorname{int} A = (0, 1)$,
- Any $x \in \mathbb{R} \setminus [0, 1]$ is an exterior point,
- $\partial A = \{0, 1\}$,
- Any $x \in [0, 1]$ is an accumulation point.

1.5.12. Example. For the set $A = \big\{x \in \mathbb{R} \colon x = \frac{1}{n}, n \in \mathbb{N} \setminus \{0\}\big\}$,

- $\operatorname{int} A = \emptyset$,
- Any $x \in \mathbb{R} \setminus (A \cup \{0\})$ is an exterior point,
- $\partial A = A \cup \{0\}$,
- Only $x = 0$ is an accumulation point.

Subsets of the Real Numbers

We can hence define general open and closed sets:

1.5.13. Definition.

(i) A set $A \subset \mathbb{R}$ is called ***open*** if all points of A are interior points, i.e., if

$$A = \operatorname{int} A.$$

(ii) A set $A \subset \mathbb{R}$ is called ***closed*** if $\mathbb{R} \setminus A$ is open.

(iii) The set

$$\overline{A} := A \cup \partial A$$

is called the ***closure*** of A. It can be shown that $\overline{A}$ is the smallest closed set that contains A.

The Complex Numbers

Although we have added the square roots (solutions to $y^2 - x = 0$, $x > 0$) to the rational numbers, and also many other elements through the addition of (P13), we have still not quite solved the problem of solutions of quadratic equations: there is no solution y to $y^2 + 1 = 0$ in the real numbers. (This is expressed by saying that $\mathbb{R}$ is ***algebraically incomplete***.)

We again need to extend the real numbers to include solutions to equations such as $y^2 = -1$. We follow the familiar procedure of considering pairs of real numbers and define the ***complex numbers*** $\mathbb{C}$ as

$$\mathbb{C} = \{(a, b) \colon a, b \in \mathbb{R}\} = \mathbb{R}^2$$

From the point of view of sets, there is no difference between $\mathbb{C}$ and $\mathbb{R}^2$. We consider the real numbers $\mathbb{R}$ as a subset of $\mathbb{C}$, writing $(x, 0) \in \mathbb{C}$ for $x \in \mathbb{R}$.

The Complex Numbers

We define addition in $\mathbb{C}$ through

$$(a_1, b_1) + (a_2, b_2) := (a_1 + a_2, b_1 + b_2),$$

so in particular the sum of two real numbers is again a real number. It is easy to verify that properties (P1)-(P4) hold for this new addition.

We next define multiplication in an apparently strange way:

$$(a_1, b_1) \cdot (a_2, b_2) := (a_1 a_2 - b_1 b_2, a_1 b_2 + a_2 b_1).$$

It turns out that this multiplication satisfies the axioms (P5)-(P8), and also (P9) together with addition. The neutral element is $(1, 0)$ (as expected), but the inverse element is a bit more complicated.

The Complex Numbers

For $z = (a, b) \in \mathbb{C} \setminus \{0\}$ the multiplicative inverse is given by

$$z^{-1} = \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right),$$

i.e., $zz^{-1} = (1, 0)$.

With the previously defined multiplication we can find a solution to the equation $z^2 = -1$:

$$(0, 1)^2 = (0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 0 \cdot 1) = (-1, 0).$$

Fundamental Theorem of Algebra

In fact, it is possible to prove that $\mathbb{C}$ is algebraically closed:

1.6.1. Fundamental Theorem of Algebra. For $a_0, \dots, a_n \in \mathbb{C}$, $a_n \neq 0$, the expression

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n$$

is said to be a ***polynomial of degree n***. There exist numbers $\alpha, \lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that

$$p(x) = \alpha \cdot (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n). \tag{1.6.1}$$

We say that the λ_k are the ***roots*** or ***zeroes*** of p. The representation (1.6.1) is called the ***factorization*** of $p(x)$.

If exactly m of the λ_k in (1.6.1) are equal to the same number $\lambda \in \mathbb{C}$, then m is said to be the ***multiplicity*** of the root λ.

We will give a proof of this result in MATH2860J next year.

The Complex Numbers

We further define multiplication of complex numbers by real numbers in the following way:

$$\lambda \cdot (a, b) = (\lambda a, \lambda b), \qquad \lambda \in \mathbb{R},\ (a, b) \in \mathbb{C}.$$

Since multiplication together with addition in $\mathbb{C}$ is distributive, we can split any complex number into two components:

$$(a, b) = (a, 0) + (0, b) = a \cdot (1, 0) + b \cdot (0, 1).$$

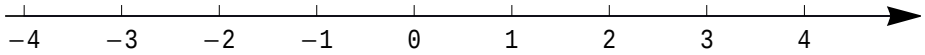
The pair $(1, 0) \in \mathbb{C}$ corresponds to $1 \in \mathbb{R}$, while the pair $(0, 1) \in \mathbb{C}$ is often denoted by the letter i. Hence we usually write

$$\mathbb{C} \ni z = (a, b) = a \cdot 1 + b \cdot i = a + bi, \qquad a, b \in \mathbb{R},$$

where $a =: \operatorname{Re} z$ is called the ***real part*** and $b =: \operatorname{Im} z$ the ***imaginary part*** of a complex number $z = a + bi$.

Visualizing Real Numbers

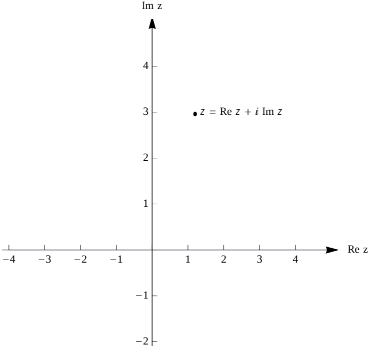
We visualize real numbers by drawing a single line, with an arrow pointing to the right, in the direction of increasing positive numbers:



We will identify the set of real numbers with this line so that a single real number corresponds to a "point" on the line. This is of course problematic; in how far this is justified, which assumptions enter into this identification and which precise axioms need to be stated falls into the purview of ***analytic geometry***. We will not discuss this further in this lecture, but simply assume a correspondence between points on a line and the real numbers.

Visualizing Complex Numbers

We visualize complex numbers by drawing two perpendicular lines ("axes"), with arrows pointing to the right and upward. each representing the real numbers. A complex number $z = a + bi = \operatorname{Re} z + (\operatorname{Im} z) \cdot i$ is represented by a point in the plane whose orthogonal projection onto the horizontal axis gives $\operatorname{Re} z$, and whose orthogonal projection onto the vertical axis gives $\operatorname{Im} z$.



The Complex Numbers

It can be shown that it is not possible to define a set P with properties (P10)-(P12) on $\mathbb{C}$; hence there exists no ordering relation "$>$" for the complex numbers.

1.6.2. Notation. Whenever we write $x > 0$, we automatically assume that $x \in \mathbb{R}$.

1.6.3. Definition. We define the ***modulus*** or ***absolute value*** of a complex number $z = a + bi$ by

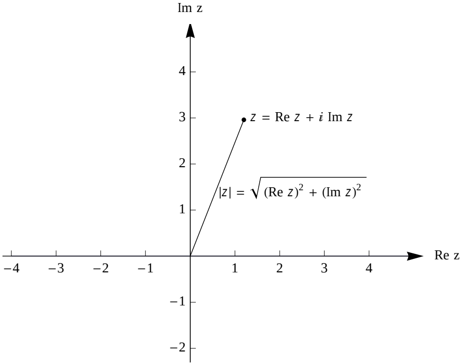
$$|z| = \sqrt{a^2 + b^2} = \sqrt{z \cdot \bar{z}}.$$

Here $\bar{z} = a - bi$ is called the ***complex conjugate*** of z.

This definition of the absolute value is called an ***extension*** of the absolute value from the real numbers to the complex numbers.

The Complex Numbers

The complex modulus has a geometric interpretation as ***distance***:



In fact, the geometric distance between two complex numbers z_1 and z_2 can is

$$\varrho(z_1, z_2) = |z_1 - z_2|.$$

The Complex Numbers

Although we cannot define the ordering $<$ on the complex numbers, we can nevertheless define the concept of bounded sets in $\mathbb{C}$.

1.6.4. Definition. Let $z_0 \in \mathbb{C}$. Then we define the ***open ball of radius $R > 0$ centered at z_0*** by

$$B_R(z_0) := \{z \in \mathbb{C} \colon |z - z_0| < R\}.$$

We say that a set $\Omega \subset \mathbb{C}$ is ***bounded*** if there exists some $R > 0$ such that $\Omega \subset B_R(0)$.

Of course, it makes no sense to talk of upper or lower bounds of sets.

The Complex Numbers

The open ball of radius R about $z_0 \in \mathbb{C}$ has a geometric interpretation as a ***neighbourhood***: